

NLP Chatbot (Question-Answering) Using Transformer

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Abstract—In the intersection of healthcare and technology, this paper introduces a cutting-edge application of NLP to forge a medical chatbot that leverages the sophisticated transformer algorithms of BERT and BioBERT. This venture is anchored in the quest to refine the delivery of medical knowledge, using AI to parse and respond to queries with unprecedented accuracy. By harnessing the specialized BioASQ dataset, this system is finely tuned to navigate the complexities of the biomedical literature, while the SQuAD dataset serves to enhance its underlying semantic comprehension. The innovation extends to the integration of Wikipedia, providing a supplementary layer of information that broadens the chatbot's scope beyond specialized medical contexts. This paper delineates the nuanced approach of transforming medical PDFs into a structured format amenable to NLP analysis, an undertaking that underscores the adaptability of transformer models to domain-specific challenges. The methodology employs strategic text chunking, enabling the chatbot to sift through data and pinpoint relevant information efficiently. The experimental outcomes of this study not only illustrate the chatbot's robust performance in medical information retrieval but also shed light on the broader implications for AI in healthcare. As the paper unfolds, it reveals the chatbot's potential to revolutionize patient care by offering a reliable, intelligent platform for medical inquiry. This research contributes a meaningful narrative to the evolving dialogue on NLP's role in enhancing the accessibility and reliability of health information systems, paving the way for future innovations in medical diagnostics and interactive patient care systems.

Index Terms—Chatbot, NLP, Transformer, BERT, BioBERT, BioASQ Dataset

I. INTRODUCTION

The integration of Natural Language Processing (NLP) into healthcare is more than a technological milestone; it is a gateway to democratizing medical knowledge. The application of NLP through intelligent chatbots has opened new horizons in medical informatics, revolutionizing the way healthcare providers and patients interact with medical data. In this landscape, this paper presents a specialized medical chatbot that utilizes the sophisticated algorithms of BERT [1] and its biomedical derivative, BioBERT [2].

The impetus for this project is rooted in the urgent need for precise, accessible, and efficient medical information dissemination. The deluge of medical literature available poses a significant challenge in information extraction and utility. Traditional search methods and generic chatbots often fall short of meeting the specific needs of the medical domain. This

paper proposes a solution: a chatbot meticulously tailored to understand and process complex medical inquiries, providing accurate and relevant information swiftly.

A key innovation of this project is the application of the transformer model, specifically BioBERT [2], which has shown exceptional promise in NLP tasks. The model's depth in understanding context and linguistic nuances is particularly beneficial in deciphering the often-complicated medical jargon. To further enhance the chatbot's capabilities, this study employs the BioASQ dataset [3], a benchmark in biomedical NLP tasks, for fine-tuning the BioBERT model.

Expanding the chatbot's versatility, we integrate Wikipedia to provide a supplementary database of information, enabling the system to cater to a broader spectrum of medical questions. Furthermore, the chatbot is designed to process text extracted from medical PDF documents, converting them into a structured format for analysis, thus leveraging the wealth of information contained in the most recent medical literature [4].

The paper elaborates on the development methodologies employed, focusing on the adaptation of the transformer models to the domain-specific datasets. It also describes the experimental setup, including the challenges of training the models on limited resources and the strategies adopted to overcome them [5]. The resultant analysis and performance evaluation of the chatbot underscore its potential as a transformative tool in medical diagnostics and patient engagement [6].

This introduction sets the stage for a comprehensive review of related literature, followed by a detailed exposition of the methodologies, experimental procedures, and analytical findings. The ultimate goal is to showcase the chatbot's role in advancing medical information retrieval and to contribute to the ongoing evolution of AI in healthcare.

II. LITERATURE WORK

The landscape of Natural Language Processing (NLP) has been continuously evolving, significantly contributing to sophisticated AI-driven applications, particularly in healthcare. This section reviews the current literature, focusing on the development of question-answering systems, the emergence

and impact of transformer models, their applications in the medical domain, and the current state of medical chatbots.

A. Question-Answering Systems in Healthcare

Automated question-answering systems, evolving from early AI's rule-based systems to recent deep learning approaches, have become crucial in healthcare for their accuracy and speed. Studies by Roberts et al. [7] and Jiang et al. [6] show the progression from simple FAQ bots to sophisticated systems capable of interpreting complex medical literature and patient data.

B. Current Medical Chatbots and Their Importance

The integration of AI-driven chatbots in healthcare, such as ChatGPT, has garnered significant attention. These chatbots are not just tools for information retrieval but also play a vital role in medical diagnostics, treatment planning, and patient engagement. The study by Chakraborty et al. [8] emphasizes the promise and application of AI-related large language models like ChatGPT in medicine, discussing their evolution, architecture, medical use, and the challenges and prospects for future research in healthcare.

C. Technologies and Challenges in Medical Chatbots

Primary technologies driving these chatbots include AI models like BERT [1] and BioBERT [2], designed for understanding complex medical queries. However, challenges persist, including ensuring data privacy, maintaining accuracy, and integrating these systems within existing healthcare frameworks.

D. The Rise of Transformer Models

The introduction of transformer models like BERT [1] and its adaptations such as BioBERT [2] has revolutionized NLP. These models excel in understanding the context and nuances of language, essential in medical applications for interpreting jargon and complex expressions.

E. Improving Medical Chatbots: Potential and Approach

While there are advancements, there is a gap in applying these technologies in user-friendly, real-world medical chatbot scenarios. This project aims to fill this gap by harnessing the power of transformer models for a practical, efficient medical chatbot. The approach involves fine-tuning BioBERT with the BioASQ dataset and integrating diverse information sources like Wikipedia and medical PDFs for a comprehensive knowledge base.

F. BioBERT and Medical Information Processing

BioBERT's adaptation for biomedical literature [2] showcases its effectiveness in medical applications. Its ability to outperform traditional models in tasks like drug-drug interaction extraction and disease-symptom relation identification signifies its potential in enhancing medical chatbots.

G. Gap in Current Literature

Despite significant advancements, there is limited exploration in practical, user-facing medical chatbots. This project contributes to bridging this gap, utilizing transformer models for an efficient and accurate medical chatbot, aligning with the evolving landscape of NLP applications in healthcare.

III. METHODOLOGY

A. Model Selection and Rationale

For this project, we carefully selected two distinct yet complementary transformer models to address different aspects of the medical question-answering task. BioBERT [2], a variant of BERT [1] specialized for biomedical contexts, was chosen for its proficiency in processing complex biomedical data. We utilized BioBERT to train the BioASQ dataset, capitalizing on its design to cater specifically to the medical setting and its capability to comprehend intricate medical terminologies and concepts.

In contrast, we employed DistilBERT, a streamlined and more computationally efficient version of BERT, for semantic training on the SQuAD dataset [5]. The decision to use DistilBERT stemmed from the need for a model that not only understands the semantic nuances of language but is also less resource-intensive compared to its larger counterparts. This choice was crucial for processing raw text data or unstructured information that is not inherently organized like the BioASQ or SQuAD datasets.

DistilBERT's role was pivotal in handling the context extracted from Wikipedia and chunks of medical PDFs. When a user poses a question, the chatbot utilizes this context, selected and processed through DistilBERT, to accurately pinpoint the answer. This approach allows the chatbot to efficiently navigate through less structured data sources, providing relevant and precise answers to user queries.

The rationale behind not using BioBERT for the entirety of the project was due to its substantial size and the extensive computational resources required for training. By adopting DistilBERT for part of the task, we aimed to strike a balance between model performance and operational feasibility, ensuring that the chatbot remains effective in diverse data scenarios while being mindful of resource constraints.

B. Data Preparation and Preprocessing

Our primary dataset for this study was a specific subset of the BioASQ dataset [3], particularly from Task B, which focuses on question-answering. Due to the complex structure of the full BioASQ dataset, we limited our scope to this single dataset, acknowledging that with additional resources, a broader range of datasets could be included. The preprocessing of the BioASQ data involved a tailored method to adapt the dataset for our chatbot's requirements.

Preprocessing of BioASQ Data: We employed a function to preprocess data that converts the BioASQ dataset into a format suitable for training our model. This process included extracting question and context texts, identifying the correct

answer span within the context, and encoding these elements into a format compatible with our model. Special attention was given to factoid-type questions where exact answers were available. The preprocessing method ensured that each data point was transformed into a structured format, comprising input IDs, attention masks, and start and end positions for the answers.

Utilizing Question-Answer Pair IDs: To make the chatbot user-friendly, we utilized the IDs of question-answer pairs from the BioASQ dataset. When a user asks a question, the system identifies the corresponding question ID based on its semantic embedding. This ID is then used to locate the relevant context within the BioASQ dataset, thereby providing the chatbot with the necessary background to generate accurate responses.

Processing PDF Files: For the PDF files, we converted the text into a raw format and then chunked it into segments. The size of these chunks was treated as a hyperparameter, allowing for experimentation to determine the optimal chunk size. These chunks served as additional context for the chatbot, enabling it to loop through them and identify relevant sections based on the user's query. The DistilBERT model [9] was then used to evaluate these sections and select the one with the highest relevance score.

Incorporating Wikipedia for Context: Wikipedia was used as an auxiliary source for context. In cases where the user's query did not match the BioASQ dataset or the PDF content, the chatbot was designed to make requests to Wikipedia, searching for context based on the keywords or phrases from the user's question. This approach allowed for broader coverage and increased the chatbot's ability to provide relevant information on a wide range of topics.

C. Training and Fine-Tuning Procedures

The fine-tuning of the BioBERT model was a crucial step in this research, conducted using the carefully processed BioASQ dataset. This process was instrumental in enhancing the model's understanding of complex medical terminologies and concepts. The training was executed on a high-performance computing setup, leveraging GPUs to handle the computational demands. We employed a masked language model approach for BioBERT, meticulously adjusting hyperparameters such as the learning rate to optimize its performance and ensure maximum efficiency and accuracy [9].

Simultaneously, we focused on refining the semantic understanding capabilities of our chatbot by fine-tuning DistilBERT on the SQuAD dataset. This step was pivotal in enabling the chatbot to interpret and process user queries with a high level of semantic accuracy. However, recognizing the resource-intensive nature of fine-tuning large language models, we adopted a strategic approach to utilize a pre-trained model as well.

To this end, we integrated the 'distilbert-base-cased-distilled-squad' model from Hugging Face's pipeline into our system. This model, already fine-tuned on the SQuAD dataset, provided a balance between performance and resource

efficiency. By employing this pre-trained model, we aimed to ensure that our chatbot remained robust in its language understanding capabilities while being mindful of the constraints related to computational resources and training time. The inclusion of the 'distilbert-base-cased-distilled-squad' model served as a valuable backup, enhancing the chatbot's performance and offering a feasible solution to the challenges posed by extensive model training.

D. Incorporating Additional Data Sources

Our approach to enriching the chatbot's knowledge base and enhancing its versatility involved integrating a variety of data sources. Medical PDFs were a primary addition, enabling the chatbot to access a wealth of up-to-date and comprehensive medical information. These PDFs provide in-depth insights into recent medical research and findings, making them invaluable for answering complex medical queries. We designed the system with scalability in mind, allowing for the inclusion of additional PDF files or documents as resources permit. This flexibility ensures that the chatbot can continually evolve and expand its knowledge base, keeping pace with advancements in medical science.

Alongside medical PDFs, we initially utilized Wikipedia as a supplementary information source. However, acknowledging the need for more reliable and up-to-date medical information, we envisage the future integration of other authoritative medical websites and databases. This shift towards more specialized and credible sources will further enhance the chatbot's ability to provide accurate and current medical information, catering to a broader spectrum of user inquiries. The potential to connect with various medical databases and websites not only broadens the chatbot's informational scope but also reinforces its utility as a dependable medical resource.

This multi-source data integration strategy ensures that our chatbot is not only versatile in handling different types of medical queries but also remains adaptable and upgradeable, reflecting our commitment to continuous improvement and alignment with the latest medical knowledge and technologies.

E. System Integration and Chatbot Framework

The fine-tuned BioBERT model was integrated into a bespoke chatbot framework developed in Python. TensorFlow was used for backend processing, ensuring that the chatbot was capable of handling and responding to user queries in real time. The chatbot's design focused on user interaction and response efficiency, ensuring a seamless and intuitive user experience.

F. Challenges and Solutions

Throughout the project, we faced challenges related to data complexity and computational resource limitations. To address these, we employed efficient data processing methods and optimized the training procedure to ensure that the chatbot could be effectively developed even with limited resources. This approach highlighted the feasibility of deploying advanced NLP models in resource-constrained environments.

IV. EXPERIMENTAL SETUP

A. Computational Resources and Challenges

The experimental phase of our project utilized the computational resources offered by Google Colab, particularly its GPU support. While this environment facilitated the training of our complex models, we encountered inherent limitations in GPU availability and computational capacity. These challenges necessitated a resource-efficient approach, shaping our experimental strategy to make the most of the available resources.

B. Model Training: BioBERT with PyTorch and DistilBERT with TensorFlow

The BioBERT model, pivotal in our experiment for biomedical text processing, was meticulously trained using PyTorch. This platform was chosen for its proficiency in handling the complex data preprocessing and fine-tuning required by BioBERT. Our training was conducted within the computational confines of Google Colab, which necessitated limiting the training to a single epoch. This was a strategic decision, balancing the model's effectiveness against the available computational resources.

For the BioASQ dataset, we selected a specific subset from its latest iteration. This decision was guided by the need to maintain a comprehensive yet manageable scope for training, considering the dataset's extensive size and the limitations of our computational setup. The preprocessing of the BioASQ dataset was adeptly managed within PyTorch, ensuring the data was optimally prepared for effective training with BioBERT.

Concurrently, our approach for the SQuAD dataset involved the DistilBERT model, for which we utilized TensorFlow. This framework was selected for its capabilities in processing and training models like DistilBERT, which required a distinct set of preprocessing and training procedures compared to BioBERT. Employing TensorFlow for DistilBERT allowed us to exploit its strengths in model training and optimization, ensuring efficient handling of the SQuAD dataset.

C. Context-ID Mapping and Semantic Search in the BioASQ Dataset

To effectively connect user queries with relevant medical information in the BioASQ dataset, we combined context-ID mapping with an advanced semantic search mechanism. This dual approach ensured precise alignment between user questions and the dataset's detailed biomedical content.

Implementing Semantic Search: For semantic search, we utilized a state-of-the-art sentence transformer model, all-MiniLM-L6-v2, from the Sentence Transformers library [10]. This model precomputed embeddings for the questions in the BioASQ dataset, enabling us to build a question-embedding dictionary. Each question ID in the dataset was mapped to its embedding, facilitating a more nuanced search based on semantic similarity.

Retrieving Context with User Queries: Upon receiving a user query, the chatbot generated its embedding using the same transformer model. It then compared this query embedding

with the precomputed question embeddings to identify the most semantically similar questions. This comparison utilized cosine similarity [9], sorting the results to extract the top matches. The chatbot then accessed the corresponding contexts for these matched question IDs from our context-ID mapping.

Enhanced Accuracy in Responses: This methodology allowed the chatbot to achieve higher response accuracy, leveraging semantic analysis to ensure relevance and alignment with the user's intent. By integrating semantic search with context-ID mapping, the chatbot could efficiently sift through the BioASQ dataset and deliver contextually and semantically appropriate responses.

System Scalability and Evolution: The incorporation of semantic search added a layer of flexibility and scalability to our system. It ensured that the chatbot could adapt to changes or expansions in the BioASQ dataset, maintaining its ability to deliver precise medical information in line with the latest developments in the field.

D. Integration with Wikipedia and Domain-Specific Sources

To augment the chatbot's versatility, we integrated Wikipedia, leveraging its extensive information repository for queries outside the BioASQ dataset. This integration involved an API-based method to query Wikipedia, retrieve relevant page titles, and extract introductory content related to user queries [11]. This approach ensures that when the BioASQ dataset lacks specific information, the chatbot can still provide valuable insights from Wikipedia.

Additionally, the chatbot was enhanced to process and utilize content from medical PDFs. Using the PDF2 library, we converted these documents into text and then segmented them into manageable chunks [12]. This chunking was performed to ensure the chatbot could efficiently navigate and extract relevant information from large text documents. Each chunk, defined by a set number of words, represented a potential context source for the chatbot's responses.

These integrations significantly expanded the chatbot's resource pool, enabling it to draw from diverse information sources for comprehensive and accurate responses. The capability to incorporate other authoritative medical websites in the future was also planned, further broadening the chatbot's scope in providing detailed and varied medical information.

E. Testing the Chatbot

The testing phase for our chatbot was designed to mimic real-world interactions, simulating a diverse array of medical queries. This involved an interactive script where users could input queries, and the chatbot would respond with information retrieved using various methods.

Interactive Chatbot Simulation: The chatbot, accessible via a simple command-line interface, prompted users to enter medical questions. Users could exit the conversation at any point by typing 'exit'. The system was equipped with functions to process user queries through different methodologies:

- 1) **BioBERT Fine-Tuned Answer:** The chatbot used the fine-tuned BioBERT model to find answers directly related to the medical context of the BioASQ dataset.

- 2) DistilBERT Fine-Tuned Answer: Queries were also processed through the fine-tuned DistilBERT model, leveraging its semantic understanding capabilities.
- 3) DistilBERT Pretrained Answer: The chatbot used a pretrained DistilBERT model ('distilbert-base-cased-distilled-squad') to handle queries by capitalizing on its inherent language comprehension.
- 4) PDF Context Answer: The chatbot processed user queries against the context obtained from chunked medical PDF text.
- 5) Wikipedia Context Answer: It utilized the Wikipedia search and fetch function to provide answers from Wikipedia for queries not covered by the BioASQ dataset.

V. RESULTS

The evaluation of our medical question-answering chatbot, which integrated BioBERT with BERT/DistilBERT models, yielded satisfactory outcomes across diverse medical queries. Despite the constraints of GPU resources on Google Colab limiting the training scope, the chatbot demonstrated a commendable ability to understand and retrieve relevant biomedical information, particularly from the BioASQ dataset.

Performance Across Different Models: BioBERT excelled in queries closely aligned with its trained dataset. Meanwhile, the integration of a BERT-based model, trained on the SQuAD dataset, enabled the chatbot to adeptly handle more generalized queries. This dual-model approach notably enhanced the chatbot's overall adaptability and response scope.

Fallback to Wikipedia and Medical Sources: The chatbot's mechanism for resorting to Wikipedia and other reliable medical sources was effective for queries beyond the primary dataset's coverage. This feature significantly contributed to maintaining the chatbot's utility across a broader range of medical inquiries.

Processing and Utilization of PDF Content: The capability to process and chunk medical PDFs expanded the chatbot's access to diverse medical knowledge. Although this feature had limitations due to its reliance on keyword-based searching, it was instrumental in providing detailed and specific answers.

User Feedback and Qualitative Assessment: User feedback, though not quantitatively analyzed, generally indicated a positive reception of the chatbot's performance. The system's versatility and ability to address a wide range of questions were particularly appreciated. Feedback also pinpointed avenues for enhancement, such as improving response naturalness and precision in context retrieval.

Evaluating the Chatbot: While we did not employ specific evaluation metrics, the chatbot's performance was qualitatively assessed based on its relevance and accuracy in responses. Future iterations could benefit from a structured evaluation using metrics like precision, recall, and F1-score, especially for comparing responses from various sources such as BioBERT, Wikipedia, PDFs, and DistilBERT.

Potential for Formal Evaluation: In future developments, formal evaluation methods, including user studies and tech-

nical assessments, could provide deeper insights into the chatbot's satisfaction rates, accuracy, and response comprehensiveness. Such evaluations would be instrumental in refining the chatbot's algorithms and enhancing its user interface and overall functionality.

In conclusion, the results highlighted the chatbot's promise as an effective tool for medical information retrieval. While not every aspect achieved optimal results, the findings were encouraging and provide a solid foundation for future advancements and refinements in this innovative project.

VI. DISCUSSION

This project's development of a medical question-answering chatbot using BioBERT and BERT/DistilBERT models has laid the groundwork for advanced applications in medical information retrieval. Despite the constraints of Google Colab's training resources, the chatbot has shown considerable promise, especially in leveraging transformer models for specialized medical inquiries.

Broadening Data Sources: Expanding the chatbot's database to include a wider array of medical information sources would enhance its accuracy and relevance. Integrating advanced transformer models like GPT or Large BERT models could also improve its query processing efficiency. To build a more sophisticated chatbot we can combine both the extractive and generative approaches as well.

User Interaction and Accessibility: Enhancing the chatbot's user interaction, possibly through natural language generation and multimodal inputs like voice recognition, could make it more user-friendly and accessible to a diverse user base.

Feedback and Evaluation: Systematic collection and analysis of user feedback, coupled with structured evaluation metrics like precision and recall, will be vital in assessing and refining the chatbot's performance. Regular user studies could provide insights into user satisfaction and areas needing improvement, informing iterative enhancements to the chatbot's design and functionality.

Leveraging Technological Advancements: Keeping pace with AI and NLP advancements will be crucial. Incorporating the latest technologies could make the chatbot more responsive and adaptive to user needs.

Overall Potential and Future Path: This project represents a significant step towards harnessing AI's potential in healthcare. Future enhancements and technological integrations hold the promise of transforming this chatbot into a more sophisticated, responsive, and reliable tool for medical professionals and patients alike. The path forward involves not only technological advancement but also a focus on user-centric design and continuous improvement based on real-world usage and feedback.

VII. CONCLUSION

This project's journey in developing a medical question-answering chatbot using transformer models like BioBERT and BERT/DistilBERT has culminated in a promising tool,

showcasing the power of NLP in healthcare. Despite computational resource limitations, the chatbot has capably processed medical queries, demonstrating the effectiveness of transformer models in specialized applications.

The chatbot's successful integration of the BioASQ dataset and Wikipedia as information sources highlights the innovative approach to broadening its knowledge base. User feedback has been encouraging, pointing to the chatbot's potential as a valuable resource for both healthcare professionals and patients, offering rapid access to medical information.

Looking to the future, the chatbot stands to benefit from extended training periods, the incorporation of more diverse and extensive datasets, and the integration of even more advanced NLP technologies. Emphasizing user feedback in its developmental iterations will further refine its accuracy and user experience.

In essence, this chatbot represents not just a successful implementation of current NLP capabilities in healthcare but also a stepping stone toward more advanced AI-assisted medical solutions. It sets a foundation for future innovations that could transform medical information retrieval, making a significant impact on healthcare delivery and accessibility.

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